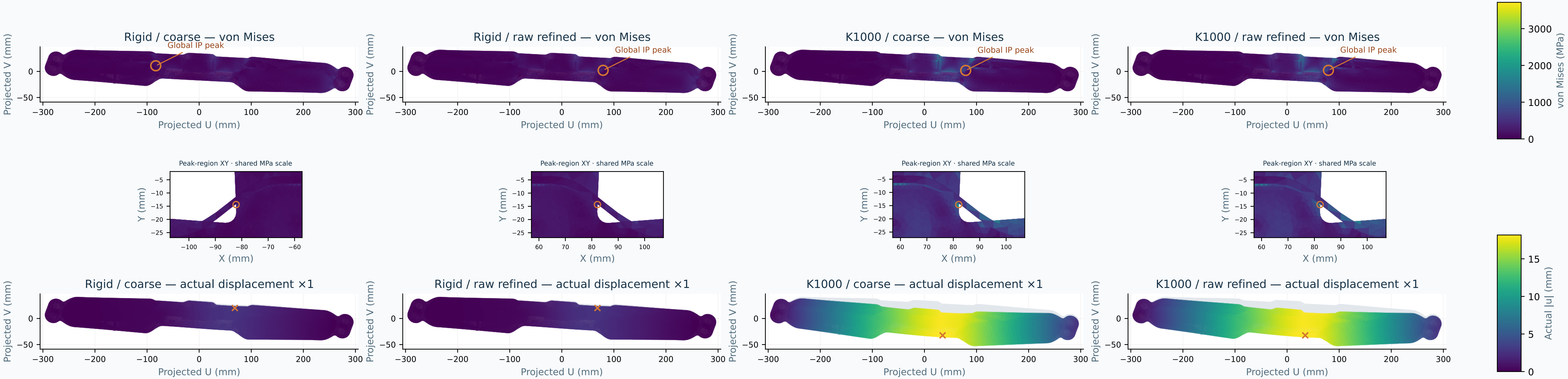
993 TURBO · PRESERVED TAPER · FOUR CLOSED LINEAR CASES

Coarse versus raw refined: stress and displacement

Fz = −4 000 N · My = +850 000 N·mm · common camera, mm axes and field scales

Coarse: 89 536 T10 / 166 403 nodes · raw refined: 91 040 T10 / 168 675 nodes · true deformation ×1 · BREP dde710a2f318...

Rigid: 4 552 nodes fixed in XYZ. K1000: 1 000 N/mm TOTAL per eye and axis, distributed over 2 274 / 2 278 nodes.



Case	Global IP peak (MPa)	Boundary max (MPa)	All-node |u| (mm)	Element / IP	Reference IP XYZ (mm)
Rigid / coarse	697.313	697.313	2.794015	84327/1	-82.13632, -14.38885, 12.86026
Rigid / raw refined	1218.072	1218.072	2.794388	89539/2	82.13018, -14.34044, 12.64617
K1000 / coarse	1977.810	1977.810	18.170201	54403/3	82.05343, -14.60357, 12.53209
K1000 / raw refined	3716.039	3716.039	18.172701	89539/2	82.13018, -14.34044, 12.64617

Stress colours: maximum of 4 IPs in each boundary-adjacent element, without nodal extrapolation. Orange circles: every global IP peak.

Linear C3D10 corner facets; actual corner displacement ×1. Grey: reference shape. Crosses: every all-node maximum of |u|.

Displacement colours use the largest of each facet’s 3 corner values; the global maximum also includes midside nodes.

Same raw BREP, two actual meshes. Assumed linear loads/material; no strength allowable or stress convergence established.

Unqualified raw taper; refined H2 volume and post-Build law refusals retained. STEP not qualified. Orthographic view 8° / 22°.